

Test Statistical methods Midterm 2025



Test ID: 297043 (In progress)

Folder: /Top/SIS/X_401020

Version: 1.9

Randomised: No

Last modified: Thursday, 25 september 2025 16:59:00

Number of questions: 16

Question order: Random

Display questions once: No

Tools: Calculator extended, Digital scratch paper

Test time: -

Cut-off: None

Maximum score: 16 pt.

Chance score: 2.55 pt. / 16%

Mark scale: -

Grading series: -

Scoring rule: No point deduction for incorrect answer

In test set with: -

Test instruction

Welcome to the exam of the course **Statistical Methods** (X_401020).

The following tools are permitted:

- calculator

Important: When you are asked to fill in a numerical value, note that a **decimal point** is used. If you use a comma instead, your answer will be considered incorrect.

For questions about the content of the test you can contact the teacher. Information about the review of the exam is provided via Canvas.

If you have **not** signed up for this exam, you will not receive a result. Through VUnet you can object to the fact that you can no longer sign up after the expiry of the registration deadline (and the fact that you will not receive a result for this exam). Submit your appeal online within one week after the exam. More information can be found at www.vu.nl/intekenen.

Welkom bij de toets van het vak **Statistical Methods** (X_401020).

Bij deze toets mag je gebruik maken van de volgende hulpmiddelen:

- rekenmachine

Belangrijk: Wanneer u een numerieke waarde moet invullen, houd er dan rekening mee dat er een **decimaal punt** wordt gebruikt. Als u in plaats daarvan een komma gebruikt, wordt uw antwoord als onjuist beschouwd.

Voor vragen over de inhoud van de toets kun je bij de docent terecht. Informatie over de toetsinzage wordt via Canvas verstrekt.

Als je **niet** hebt ingetekend voor dit tentamen, ontvang je geen resultaat. Via VUnet kun je bezwaar maken tegen het feit dat je niet meer kunt intekenen na het verstrijken van de intekendeadline (en je dus geen resultaat krijgt voor dit tentamen). Dien je bezwaarschrift online in binnen één week na het tentamen. Meer informatie vind je op www.vu.nl/intekenen.

Question 1 – 2_sets_probability – 505622.1.1

Question type: Multiple response

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: Yes

Maximum score: 1

Chance score: 0.33 pt. / 33%

Status:

Last modified: 25/09/2025 17:02

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Give all correct answers: $P(A \cap B) - P(A^c \cap B^c) = \dots$

- A** $1 - P(A) - P(B)$
- B** $P(A) + P(B) - 1$
- C** $1 - P(A^c) - P(B^c)$
- D** $P(A^c) + P(B^c) - 1$

Feedback at question level

$P(A^c \cap B^c) = 1 - P(A \cup B)$, $P(A \cup B) + P(A \cap B) = P(A) + P(B)$

Question 2 – bayes – 505670.2.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 17:18

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Suppose that 2% of all produced engines are broken. If an engine is broken, the final test is positive with a probability of 97%. If the engine is not broken, the final test is negative with a probability of 99%. Knowing that the final test is positive, what is the probability that corresponding engine is actually functioning well? Round to two decimals.

0.34

Feedback at question level

For $A = \{\text{engine functioning well}\}$, $B = \{\text{positive test}\}$ we get $P(A) = 0.98$, $P(B | A) = 0.01$, $P(B | A^c) = 0.97$ and hence $P(A | B) = (0.01 * 0.98) / (0.01 * 0.98 + 0.97 * 0.02)$.

Question 3 – Bernoulli_sum – 505712.3.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.25 pt. / 25%

Status:

Last modified: 05/09/2025 16:28

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that X and Y are two independent Bernoulli random variables with parameter $p=0.3$. Then the random variable $X+Y$ has the following distribution:

- A** Bernoulli
- B** geometric
- ☒ **C** binomial
- D** neither of the above

Question 4 – binomial – 505702.1.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 17:21

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

A person is stepping on a bus which is filled with 5 passengers. Assume that the probability that a single passenger is wearing sneakers is 20%. What is the probability that precisely 2 of them are wearing sneakers? Round to three decimals.

0.205

Feedback at question level

$X \sim \text{Binomial}(n,p)$ with $n=5$, $p=0.2$.

Question 5 – binomial_Poisson – 505735.1.2

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.33 pt. / 33%

Status:

Last modified: 28/08/2025 12:47

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that $X \sim \text{Binomial}(1000, 0.001)$. Then $P_Y(5)$ gives a good approximation to $P_X(5)$ for

- A** $Y \sim \text{Poisson}(1)$
- B** $Y \sim \text{Pascal}(1000, 0.001)$
- C** $Y \sim \text{Binomial}(100, 0.001)$

Question 6 – Cartesian_product – 505561.1.3

Question type: Kprime statement

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: Yes

Maximum score: 1

Chance score: 0.50 pt. / 50%

Status:

Last modified: 28/08/2025 16:25

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Indicate for each statement about the Cartesian product of sets whether it is true or false.

	True	Not true
A and B are countable $\Rightarrow A \times B$ is countable	X	
$A \times B = B \times A$ for all sets A, B		X

Question 7 – disjoint_sets – 505559.2.0

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.25 pt. / 25%

Status:

Last modified: 05/09/2025 16:33

Attributes: Taxonomie/Taxonomy 00 Onbekend/Unknown

Three sets A, B, and C are called disjoint if

- A** $A \cap B \cap C = \emptyset$
- B** $A \cap B = \emptyset$ and $B \cap C = \emptyset$ and $C \cap A = \emptyset$
- C** $A \cap B = \emptyset$ or $B \cap C = \emptyset$ or $C \cap A = \emptyset$
- D** neither of the above

Question 8 – disjoint+independent – 505624.1.1

Question type: Kprime statement

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: Yes

Maximum score: 1

Chance score: 0.19 pt. / 19%

Status:

Last modified: 25/09/2025 17:58

Attributes: Taxonomie/Taxonomy 00 Onbekend/Unknown

The two sets A and B are disjoint and independent. What is always true?

	True	Not true
$A=\emptyset$ and $B=\emptyset$		X
$A=\emptyset$ or $B=\emptyset$		X
$P(A)=0$ or $P(B)=0$	X	
$P(A)=0$ and $P(B)=0$		X

Feedback at question level

$P(A) * P(B) = P(A \cap B) = P(\emptyset) = 0$

Question 9 – geometric_compute – 505699.1.3

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:02

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that a random variable is geometrically distributed with parameter $p = 1/4$. What is $P_X(3)$? Round to two decimals.

0.14

Feedback at question level

$$P_X(3) = 1/4 * (1 - 1/4)^2 = 9/64$$

Question 10 – independent – 505671.1.3

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.50 pt. / 50%

Status:

Last modified: 25/09/2025 18:06

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

One randomly chooses a number from the set $S = \{1, \dots, 20\}$. Consider $A = \{1, 4, 9, 16\}$ and $B = \{3, \dots, 12\}$. Are A and B independent?

☒ **A** yes

☐ **B** no

Feedback at question level

With $A \cap B = \{4, 9\}$ we get $P(A) * P(B) = 4/20 * 10/20 = 2/20 = P(A \cap B)$.

Question 11 – Pascal_compute_multiple – 505730.2.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.20 pt. / 20%

Status:

Last modified: 25/09/2025 18:11

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

A random variable X is negatively binomially distributed, $X \sim \text{Pascal}(2, 0.3)$. What is $P_X(5)$, rounded to two decimals ?

A 0.12

B 0.71

C 0.33

D 0.51

E 0.03

Feedback at question level

$$P_X(5) = (5 - 1) * (0.3)^2 * (1 - 0.3)^{5-2}$$

Question 12 – PMF_die – 505673.2.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:17

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that one throws a dice twice. Let X be the random variable given by the sum of the two results. What is the value of the probability mass function at $x=5$? Round to two decimals.

0.11

Feedback at question level

$$P_X(5) = P(X = 5) = P(A_5) = \frac{4}{36} \text{ since } A_5 = \{(1,4), (2,3), (3,2), (4,1)\}$$

Question 13 – Poisson – 505732.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:21

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

On average, 2 people use the bus from Amsterdam Zuid to Haarlem which departs at 11pm. Using the Poisson distribution, what is the probability that there are precisely 2 passengers on today's ride? Round to two decimals.

0.27

Feedback at question level

$X \sim \text{Poisson}(2)$ and hence $P_X(2) = (e^{-2} * 2^2) / (2!)$

Question 14 – same_choice – 505739.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:23

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Four people have been asked to think of a number between 1 and 10. What is the probability that everyone is thinking of a different number? Round to one decimal.

0.5

Feedback at question level

$10/10 * 9/10 * 8/10 * 7/10$

Question 15 – union_independent – 505667.2.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:25

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that $A, B \subset S$ are independent with $P(A)=\frac{2}{3}$ and $P(B)=\frac{1}{4}$. What is $P(A \cup B)$ (round to two decimals)?

0.75

Feedback at question level

$P(A \cup B) = P(A) + P(B) - (P(A) * P(B))$

Question 16 – variable_conditional – 505678.2.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Midterm 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 25/09/2025 18:30

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

A dice is rolled three times. Let X_i denote the result of the i^{th} role for $i=1,2,3$. What is $P(X_1+X_2=5, X_2+X_3=4 \mid X_1=3)$? Round to three decimals.

0.028

Feedback at question level

$P(X_1+X_2=5, X_2+X_3=4 \mid X_1=3) = P(X_2=2, X_3=2) = \frac{1}{6} * \frac{1}{6}$

Test matrix

No matrix defined